MODULE 6.2

ZPF Operation

Zone-Based Policy Firewalls | Transit and Self-Zone Traffic

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Cisco Networking Academy | 2026

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ZPF Actions

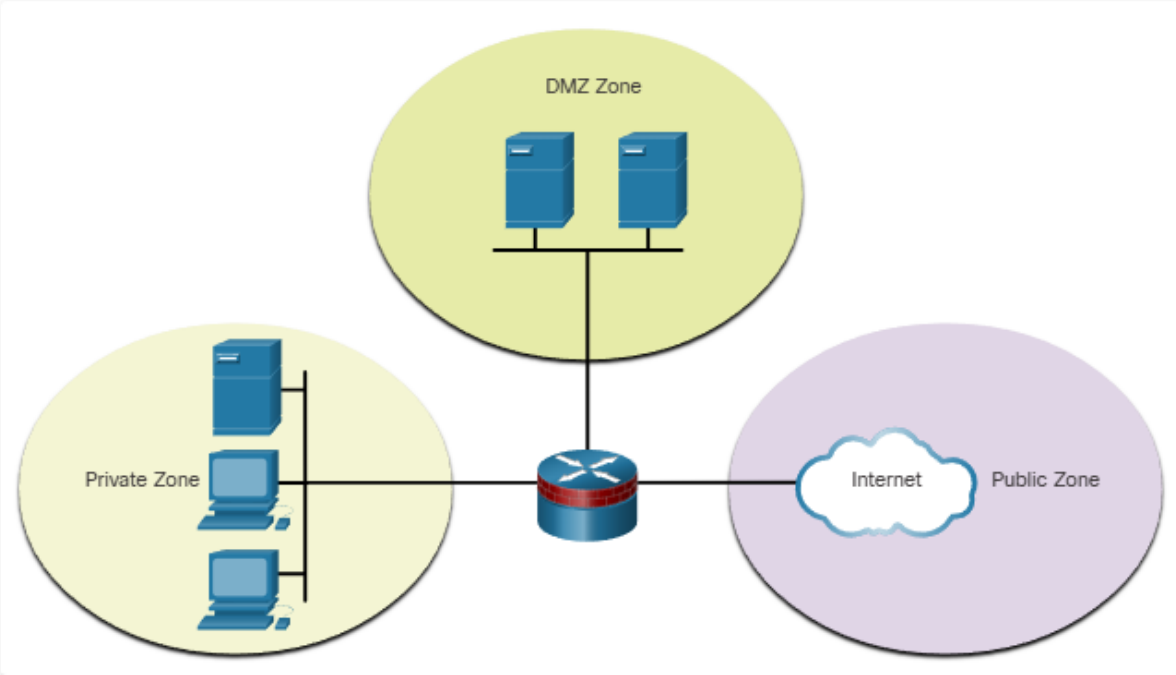
A zone-based policy firewall can apply exactly one of three actions to traffic crossing a zone pair:

Action	What it does	When it's used
Inspect	Performs stateful packet inspection — the firewall tracks the connection's state and automatically permits return traffic for sessions that originated from the zone.	The default choice for most zone pairs that need normal, bidirectional application traffic.
Drop	Silently discards traffic that matches the policy (an optional "log" keyword records the match) — functionally the same as a deny statement in an ACL.	The implicit default for any zone pair that has no explicit policy configured.
Pass	Allows traffic through with no stateful tracking at all — functionally the same as a permit statement in an ACL.	Stateless or exempted traffic; because it isn't stateful, it must be configured in both directions if traffic needs to flow both ways.

Abbreviations: ZPF = Zone-Based Policy Firewall. ACL = Access Control List.

Rules for Transit Traffic

- **Depends on zone membership:** the rules depend on whether the ingress and egress interfaces are members of the same zone.
- **Neither interface is a zone member:** the resulting action is to pass the traffic.
- **Both interfaces are members of the same zone:** the resulting action is to pass the traffic — same-zone traffic is unrestricted by default.
- **One interface is a zone member, the other is not:** the resulting action is to drop the traffic, regardless of whether a zone-pair exists.
- **Both interfaces belong to the same zone-pair and a policy exists:** the resulting action is inspect, allow, or drop, exactly as defined by that policy.



Traffic transiting through router interfaces is subject to these rules. The example above uses this Basic Security Zone Topology.

Rules for Transit Traffic: Summary Table

Source Interface Member of Zone?	Destination Interface Member of Zone?	Zone-Pair Exists?	Policy Exists?	Result
NO	NO	N/A	N/A	PASS
YES	NO	N/A	N/A	DROP
NO	YES	N/A	N/A	DROP
YES (private)	YES (private)	N/A	N/A	PASS
YES (private)	YES (public)	NO	N/A	DROP
YES (private)	YES (public)	YES	NO	DROP
YES (private)	YES (public)	YES	YES	INSPECT

The table summarizes the four rules from the previous slide for the private/DMZ/public topology.

Rules for Traffic to the Self Zone

- **The self zone:** the router itself, including every IP address assigned to its interfaces — traffic that originates at the router or is addressed to a router interface, such as SSH management traffic or routing protocol traffic.
- **Different rules:** the rules for a ZPF are different for the self zone than for transit traffic.
- **Default: permit:** if the router is the source or the destination of the traffic, all traffic is permitted by default.
- **The one exception:** if the source and destination form a zone-pair with a specific service-policy configured, that policy is applied to all the traffic instead.

Source Interface Member of Zone?	Destination Interface Member of Zone?	Zone-Pair Exists?	Policy Exists?	Result
YES (self zone)	YES	NO	N/A	**PASS**
YES (self zone)	YES	YES	NO	**PASS**
YES (self zone)	YES	YES	YES	**INSPECT**
YES	YES (self zone)	NO	N/A	**PASS**
YES	YES (self zone)	YES	NO	**PASS**
YES	YES (self zone)	YES	YES	**INSPECT**

Check Your Understanding

Given the conditions, what's the resulting action?

Q: Neither the source nor the destination interface is a zone member.

A: PASS

Q: The source interface is a zone member (private); the destination is not a zone member.

A: DROP

Q: Both interfaces are members of the same zone-pair, a zone-pair exists, but no policy has been configured.

A: DROP

Q: Both interfaces are members of the same zone-pair, a zone-pair exists, and a policy is configured.

A: INSPECT (per the policy)

Q: Traffic originates from the self zone; no zone-pair exists for the destination.

A: PASS

Q: Traffic originates from the self zone; a zone-pair exists and a policy is configured.

A: INSPECT

Summary

- **6.2.1 —** a ZPF applies exactly one of three actions to traffic crossing a zone pair: inspect (stateful, auto-permits return traffic), drop (the default for any zone pair with no policy), or pass (stateless, must be configured both ways).

- **6.2.2 —** for transit traffic, same-zone and no-zone-membership traffic passes by default; traffic crossing from a zone member to a non-member is always dropped; only traffic crossing a configured zone-pair follows the policy.

- **6.2.3 —** the self zone (the router itself) is the one exception — traffic to or from the router is permitted by default, unless a specific zone-pair policy says otherwise.

- **6.2.4 —** every scenario reduces to the same two tables — knowing them cold means never having to guess whether a rule will pass, drop, or inspect.

References

Cisco Networking Academy — Network Defense, Module 6.2: ZPF Operation

https://www.netacad.com

Cisco — Zone-Based Policy Firewall Design and Application Guide

https://www.cisco.com/c/en/us/support/docs/security/ios-firewall/98628-zone-design-guide.html

Cisco — Zone-Based Firewall High Availability

https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/sec_data_zbf/configuration/xe-3s/sec-zone-pol-fw-xe-3s-book.html